

01 Data & Frequency Distribution

- Data may be **raw** (ungrouped) or a **frequency distribution** (grouped into classes).
- Class mark $x_i = \frac{\text{lower} + \text{upper}}{2}$; class width h ; frequency f_i ; $N = \sum f_i$.
- Cumulative frequency (cf): "less-than" and "more-than" running totals.

02 Central Tendency

"Averages" describing the centre of data:

- Mean** (arithmetic average), **Median** (positional middle), **Mode** (most frequent).
- Special means: **Geometric** (GM) and **Harmonic** (HM).

03 Arithmetic Mean

- Direct: $\bar{x} = \frac{\sum f_i x_i}{N}$.
- Assumed mean: $\bar{x} = A + \frac{\sum f_i d_i}{N}$, $d_i = x_i - A$.
- Step-deviation:
 $\bar{x} = A + h \cdot \frac{\sum f_i u_i}{N}$, $u_i = \frac{x_i - A}{h}$.
- $\sum f_i (x_i - \bar{x}) = 0$.

04 Weighted & Combined Mean

- Weighted mean: $\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$.
- Combined mean of two groups:
 $\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$.

05 Median

- Ungrouped (n ordered): odd $\Rightarrow \left(\frac{n+1}{2}\right)$ th term; even $\Rightarrow$ mean of $\frac{n}{2}$ th & $\left(\frac{n}{2} + 1\right)$ th.
- Grouped: $\text{Md} = \ell + \frac{\frac{N}{2} - C}{f} \times h$.

ℓ, C, f, h = lower limit, preceding cf, frequency, width of median class.

06 Mode

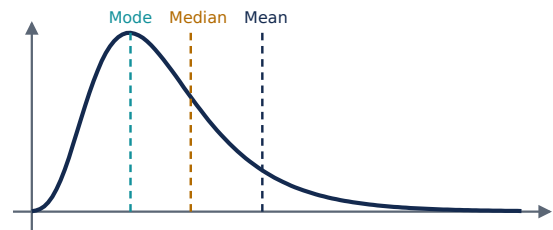
$$\text{Mo} = \ell + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

f_1 = modal freq.; f_0, f_2 = freq. of preceding/following class; ℓ, h as usual.

07 Geometric & Harmonic Mean

- $\text{GM} = (x_1 x_2 \cdots x_n)^{1/n}$, $\log \text{GM} = \frac{1}{n} \sum \log x_i$.
- $\text{HM} = \frac{n}{\sum \frac{1}{x_i}}$.
- Inequality:** $\text{AM} \geq \text{GM} \geq \text{HM}$ (equality iff all values equal).
- Two numbers: $\text{GM}^2 = \text{AM} \times \text{HM}$.

08 Empirical Relation & Skew



Right-skewed: Mode < Median < Mean

- Moderately skewed: **Mode = 3 Median - 2 Mean**.
- Symmetric: Mean = Median = Mode. Right skew: Mode < Md < Mean.

09 Measures of Dispersion

- Absolute:** range, mean deviation, standard deviation.
- Relative** (coefficients): unit-free ratios used to compare data sets.

10 Range & Coefficient

- Range = $L - S$ (largest - smallest).
- Coefficient of range = $\frac{L - S}{L + S}$.
- Simplest measure; uses only the two extreme values.

11 Mean Deviation

- About a (mean or median): $\text{MD} = \frac{\sum f_i |x_i - a|}{N}$.
- MD is **least when taken about the median**.
- Coefficient of MD = $\frac{\text{MD}}{a}$.

12 Variance & Standard Deviation

$$\sigma^2 = \frac{1}{N} \sum f_i(x_i - \bar{x})^2, \quad \sigma = \sqrt{\sigma^2}.$$

$$\sigma^2 = \frac{\sum f_i x_i^2}{N} - \left(\frac{\sum f_i x_i}{N} \right)^2$$

$$\text{Step-deviation: } \sigma^2 = h^2 \left[\frac{\sum f_i u_i^2}{N} - \left(\frac{\sum f_i u_i}{N} \right)^2 \right]$$

13 RMS & Minimisation

- $\text{RMS}^2 = \frac{\sum f_i x_i^2}{N}$; $\sigma^2 = \text{RMS}^2 - \bar{x}^2$.
- $\sum (x_i - a)^2$ is **minimum** at $a = \bar{x}$.
- $\sum |x_i - a|$ is minimum at $a = \text{median}$.

14 Change of Origin & Scale

If $y_i = a + h x_i$:

Measure	Effect
Mean	$\bar{y} = a + h\bar{x}$
Variance	$\sigma_y^2 = h^2 \sigma_x^2$
Std. deviation	$\sigma_y = h \sigma_x$

Dispersion is unchanged by a shift of origin.

15 Combined Variance

Groups $(n_1, \bar{x}_1, \sigma_1)$ and $(n_2, \bar{x}_2, \sigma_2)$:

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$$

$d_1 = \bar{x}_1 - \bar{x}$, $d_2 = \bar{x}_2 - \bar{x}$; $\bar{x}$ = combined mean.

16 Coefficient of Variation

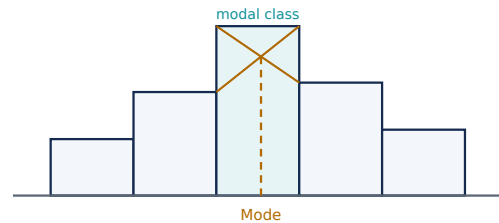
- $\text{CV} = \frac{\sigma}{\bar{x}} \times 100\%$ (**unit-free** relative spread).
- Smaller CV $\Rightarrow$ more **consistent** data.
- Use CV — not raw σ — to compare sets with different means/units.

17 Standard Results

For $1, 2, \dots, n$:

- Mean = $\frac{n+1}{2}$; Variance = $\frac{n^2-1}{12}$.
- $\sum i = \frac{n(n+1)}{2}$, $\sum i^2 = \frac{n(n+1)(2n+1)}{6}$.

18 Graphical Mode — Histogram



- Tallest rectangle = modal class.
- Cross-join the modal bar's top corners to the adjacent bars; the intersection dropped to the axis gives the **mode**.

19 Comparing Distributions

- Same mean, larger $\sigma \Rightarrow$ more spread / less consistent.
- Compare consistency using CV (unit-free).

20 Quick Recap

- $\sigma^2 = \frac{\sum f x^2}{N} - \bar{x}^2$; $\text{AM} \geq \text{GM} \geq \text{HM}$.
- Mode = $3\text{Md} - 2\text{Mean}$ (moderately skewed).
- $y = a + hx \Rightarrow \bar{y} = a + h\bar{x}$, $\sigma_y = |h|\sigma_x$.
- $\text{CV} = \frac{\sigma}{\bar{x}} \times 100$; smaller = more consistent.
- 1..n: variance = $\frac{n^2-1}{12}$.

21 Common Traps

- ! Variance uses the **mean**; MD may use mean or median, but MD is least about the median.
- ! Shift of origin leaves σ unchanged; scaling by h gives $\sigma_y = |h|\sigma_x$, variance $\times h^2$.
- ! In combined variance, never drop the d_i^2 terms.
- ! $\text{AM} \geq \text{GM} \geq \text{HM}$ holds only for **positive** values.
- ! Compare consistency by CV, not raw σ ; and $\sigma^2 \geq 0$ always.